

Real-time Pacing



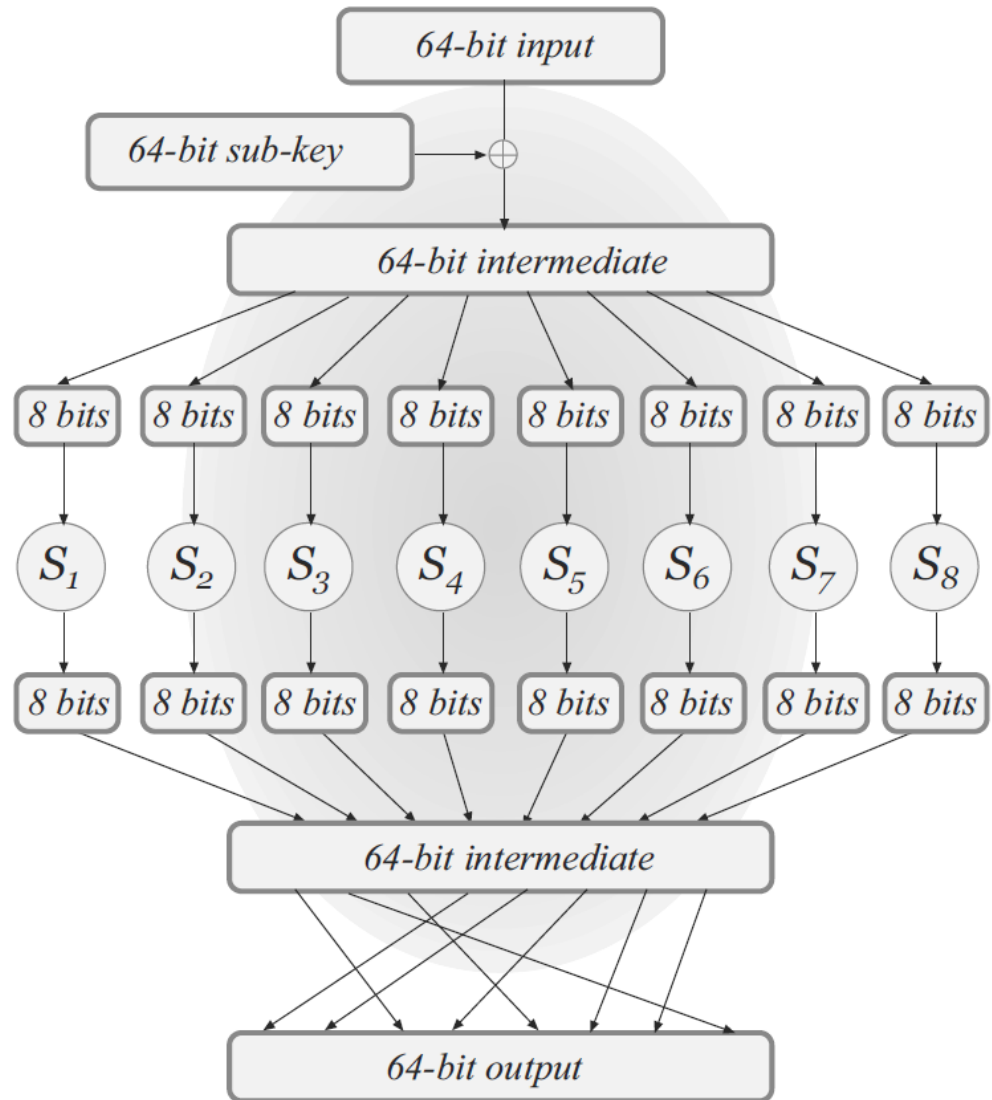
<https://forms.gle/u8FR8MZg8sLPVrmM8>

Leading Questions

How to construct a real-world CPA-secure encryption scheme from block cipher?

1. What are the constructions of block cipher?
2. What are the encryption modes?
3. Security/Attacks?

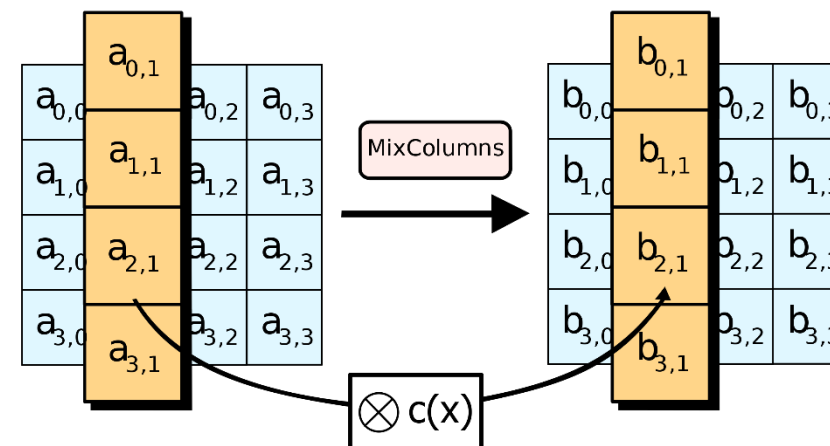
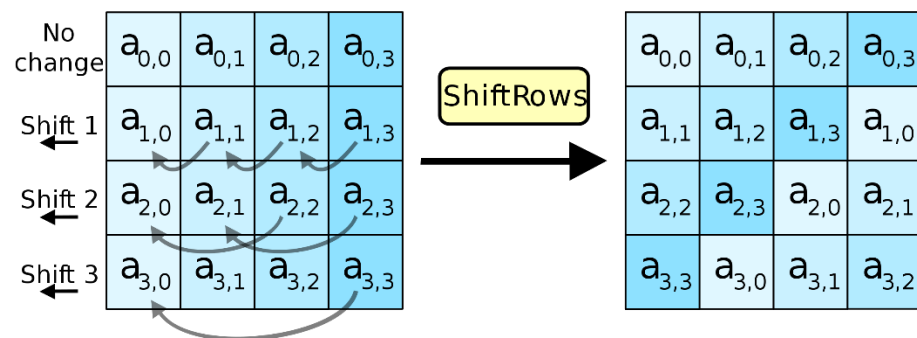
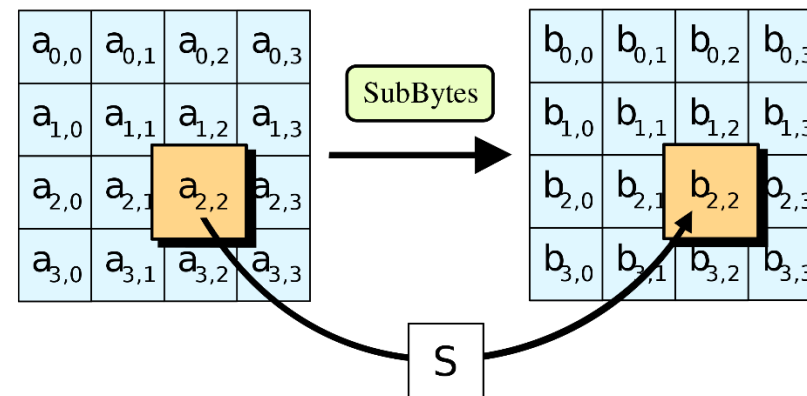
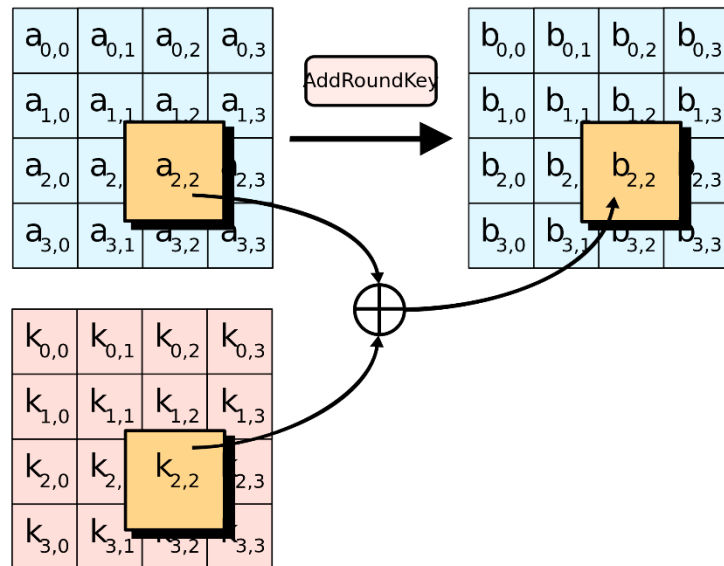
Substitution-Permutation Networks (SPN)



1. XOR with secret key
2. Substitution boxes
3. Permutation

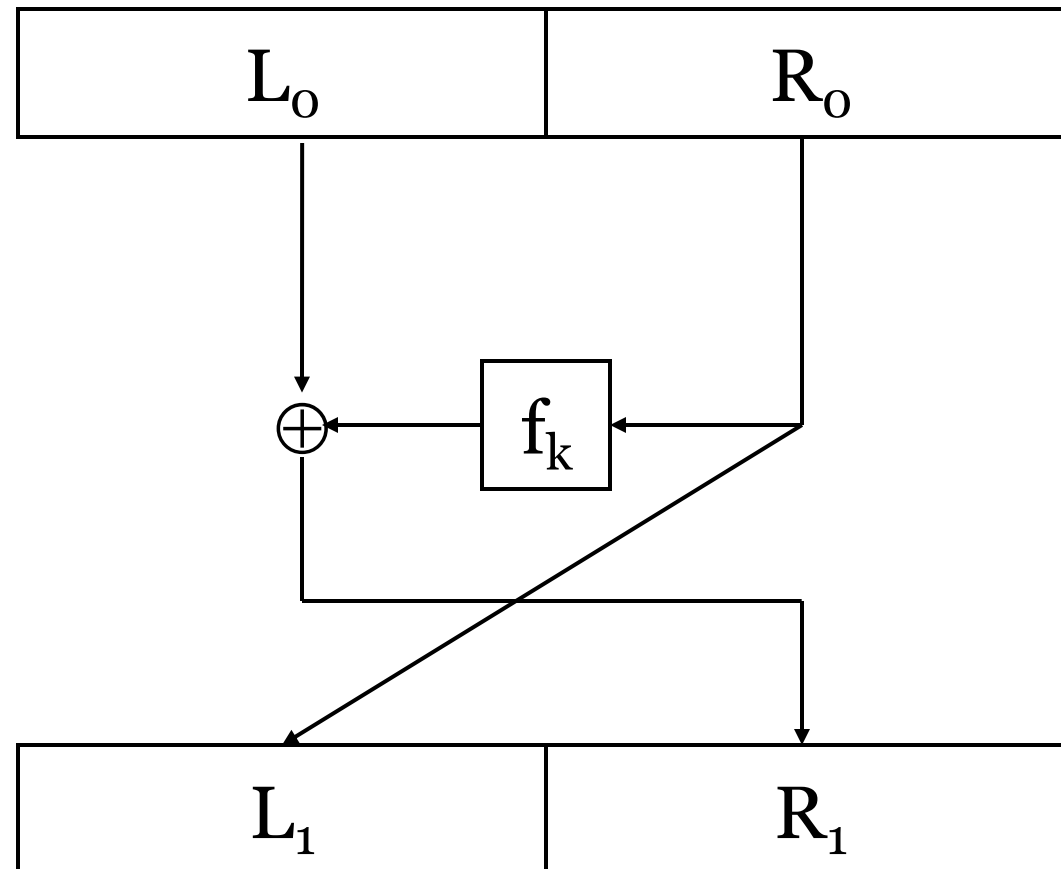
AES

https://en.wikipedia.org/wiki/Advanced_Encryption_Standard

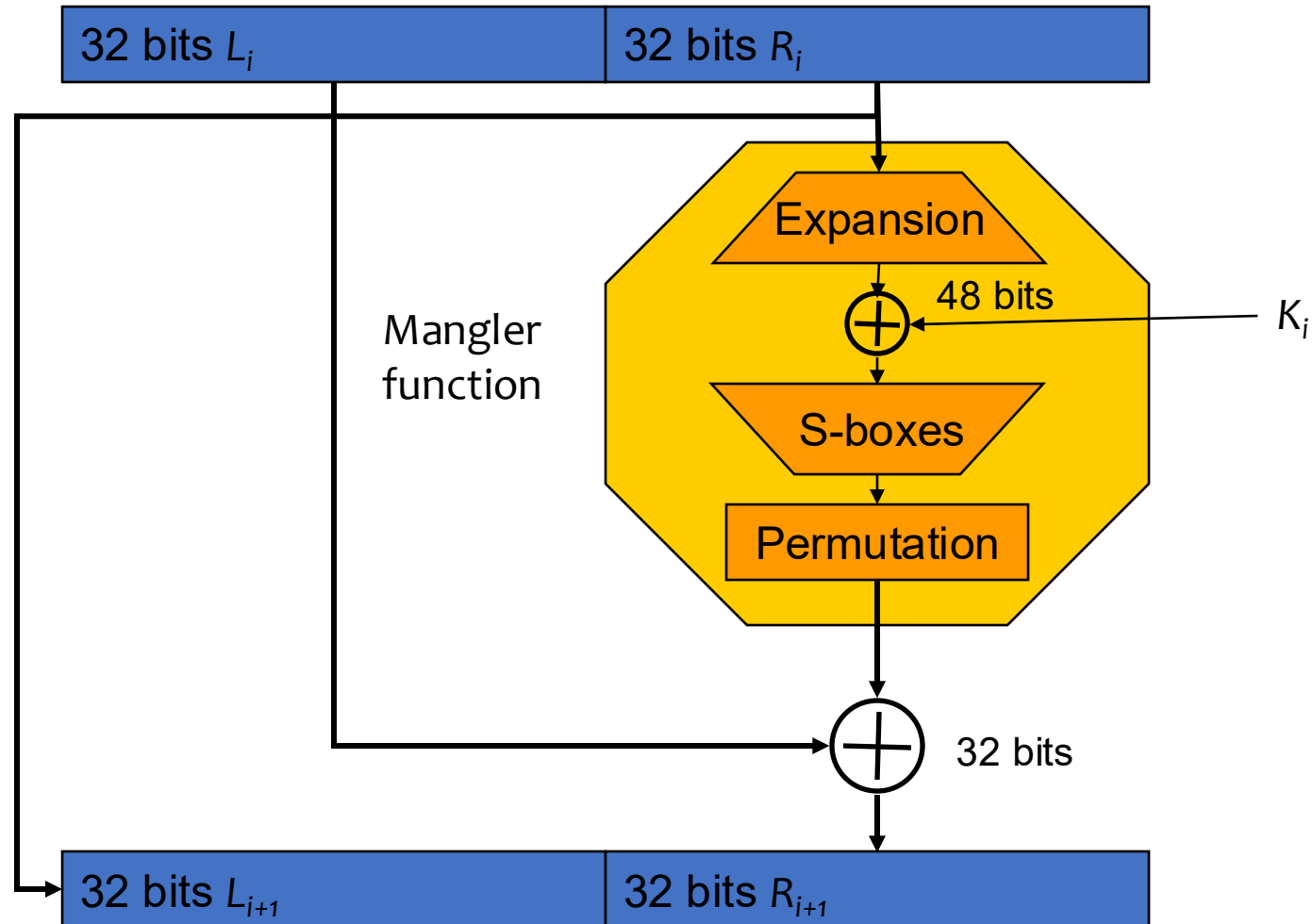


Feistel network

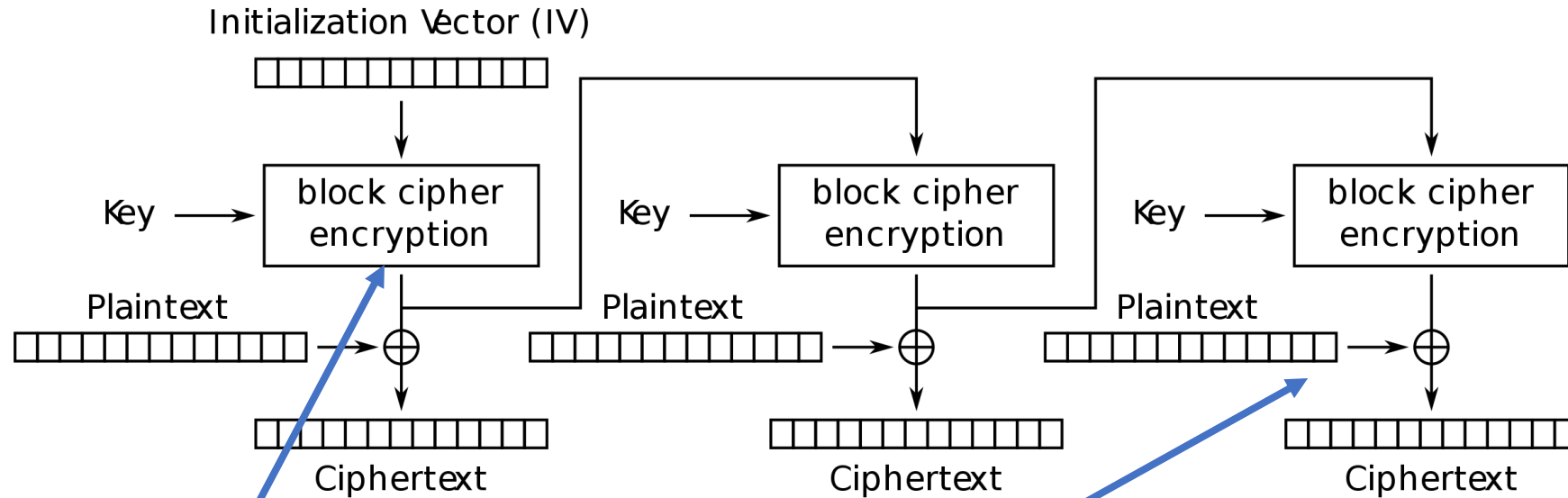
$$L_i = R_{i-1}, R_i = L_{i-1} \oplus f_i(R_{i-1})$$



DES



Recall

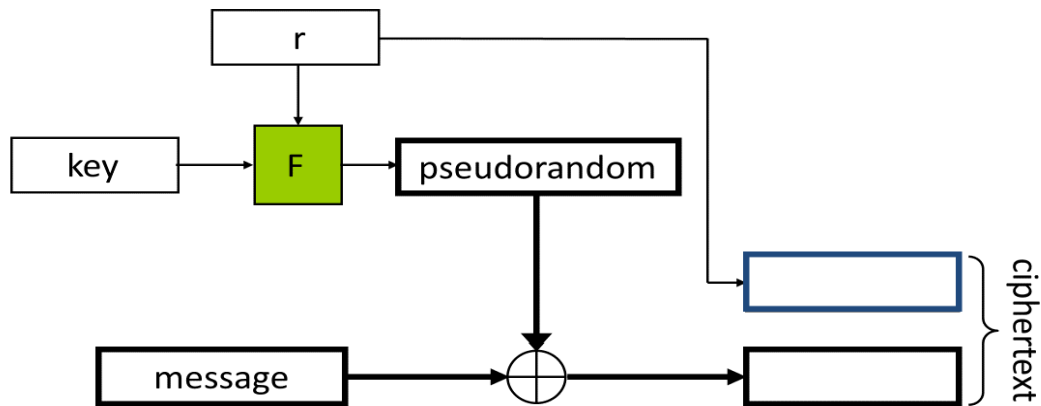


Output Feedback (OFB) mode encryption

1. Block cipher
2. Mode of encryption
3. Padding

Today: Block Cipher Modes

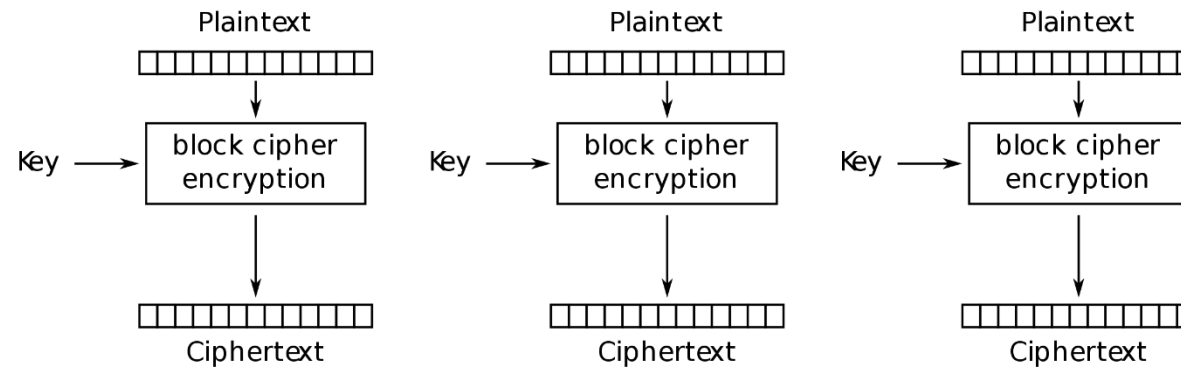
- A block cipher mode describes the way a block cipher encrypts and decrypts a sequence of message blocks.



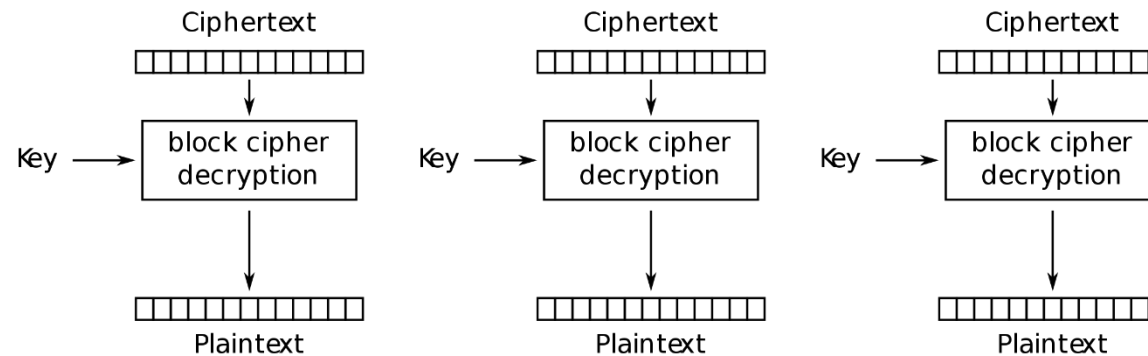
ECB mode

Block $M[i]$ encrypted into ciphertext block $C[i] = \text{Enc}_k(M[i])$

Block $C[i]$ decrypted into plaintext block $M[i] = \text{Dec}_k(C[i])$



Electronic Codebook (ECB) mode encryption



Electronic Codebook (ECB) mode decryption

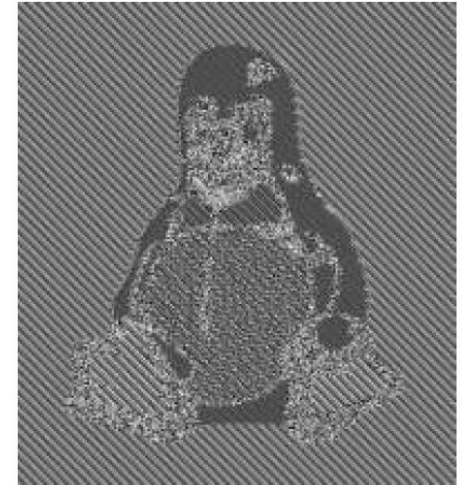
Insecurity of ECB

- Property:
 - Identical plain text blocks are encrypted to identical cipher text blocks
- Weakness:
 - Documents and images are not suitable for ECB encryption since patterns in the plaintext are repeated in the ciphertext:

What do we need to avoid repeatable patterns in ciphertext?



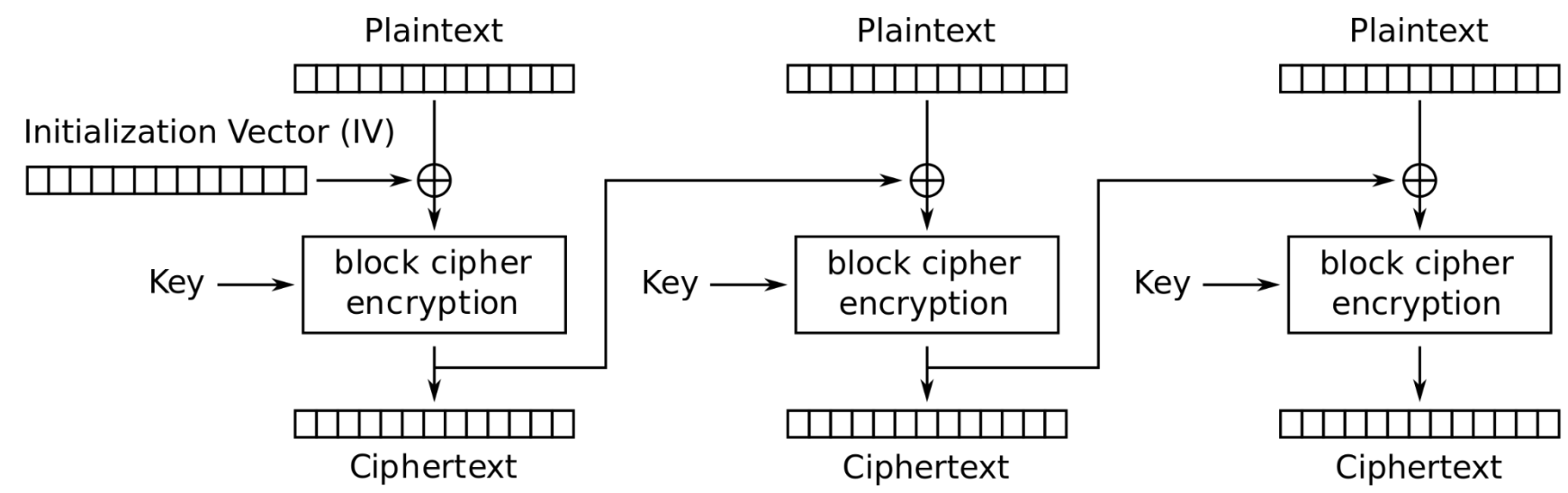
(a)



(b)

Figure 8.6: How ECB mode can leave identifiable patterns in a sequence of blocks: (a) An image of Tux the penguin, the Linux mascot. (b) An encryption of the Tux image using ECB mode. (The image in (a) is by Larry Ewing, lewing@isc.tamu.edu, using The Gimp; the image in (b) is by Dr. Juzam. Both are used with permission via attribution.)

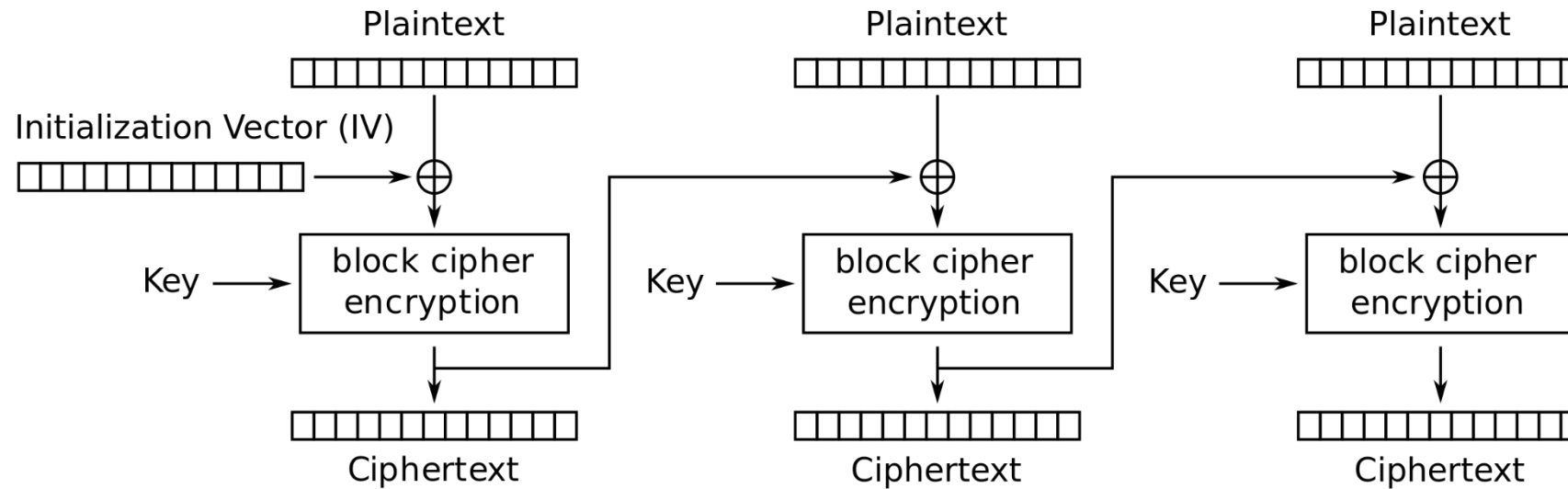
Cipher Block Chaining (CBC) Mode



Cipher Block Chaining (CBC) mode encryption

Cipher Block Chaining (CBC) Mode

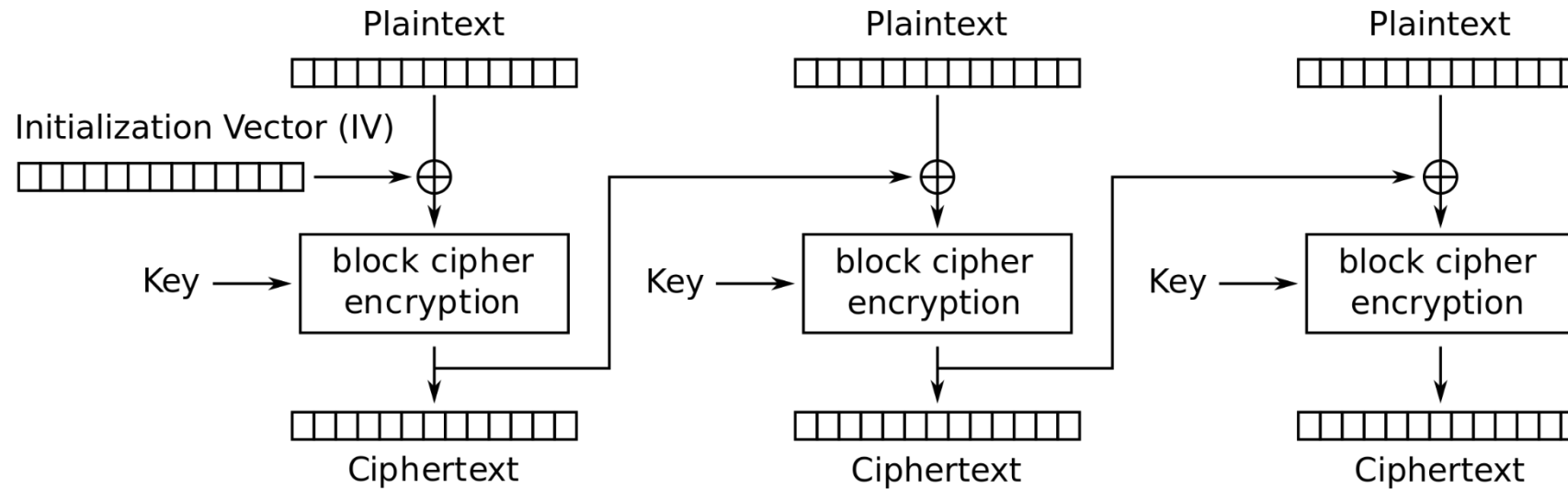
- The previous ciphertext block is combined with the current plaintext block $C[i] = \text{Enc}_K(C[i-1] \oplus M[i])$
- $C[-1] = \text{IV}$, a random block separately transmitted encrypted (known as the initialization vector, randomly chose every time)
- Decryption: $M[i] = C[i-1] \oplus \text{Dec}_K(C[i])$



Cipher Block Chaining (CBC) mode encryption

Cipher Block Chaining (CBC) Mode

- The previous ciphertext block is combined with the current plaintext block $C[i] = \text{Enc}_K(C[i-1] \oplus M[i])$
- $C[-1] = \text{IV}$, a random block separately transmitted encrypted (known as the initialization vector, randomly chose every time)
- Decryption: $M[i] = C[i-1] \oplus \text{Dec}_K(C[i])$



Cipher Block Chaining (CBC) mode encryption

Reduction of block cipher

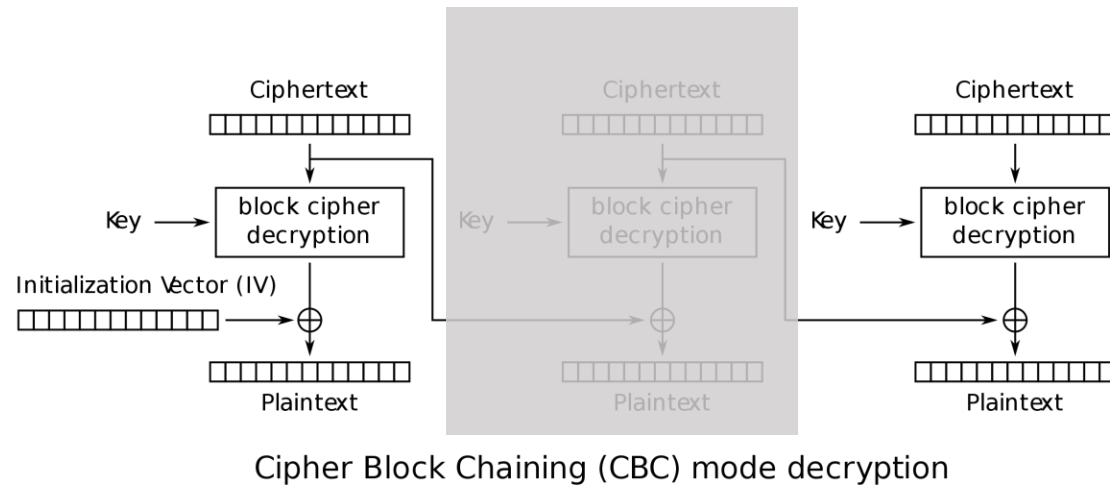
- Assumption: Pseudo-Random Permutation
- Prove security of block cipher modes
 - Very non-trivial

CRYPTO 2019 Best Paper

Cryptanalysis of OCB2: Attacks on Authenticity and Confidentiality, Akiko Inoue, Tetsu Iwata, Kazuhiko Minematsu, Bertram Poettering

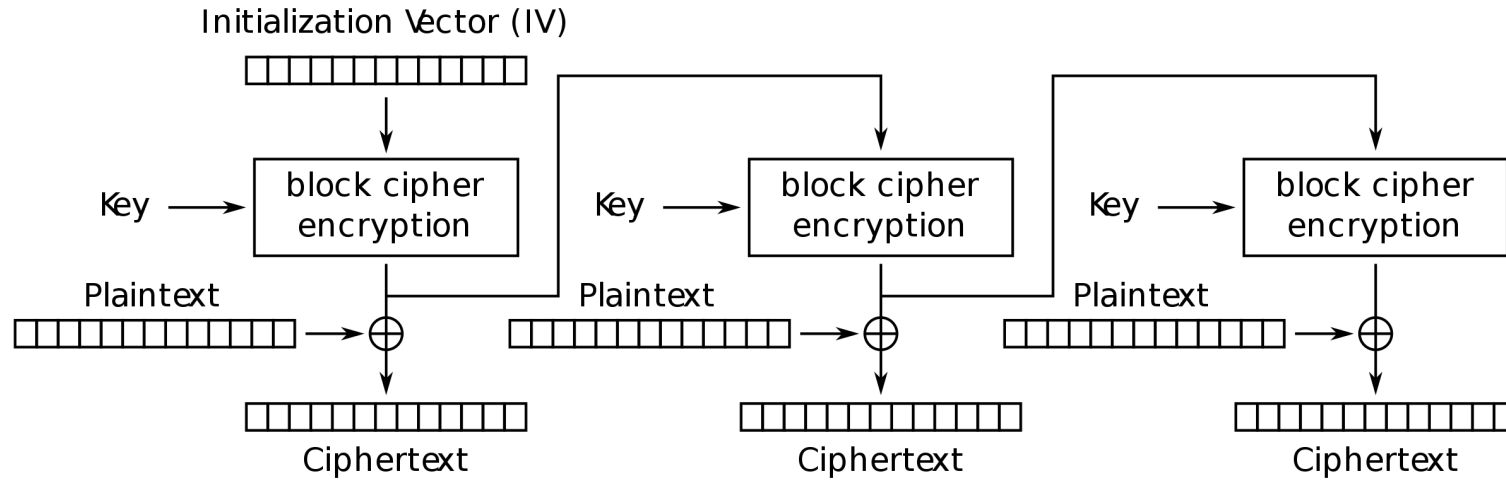
Strengths and Weaknesses of CBC

- Strengths:
 - CPA-secure
 - Is fast and relatively simple

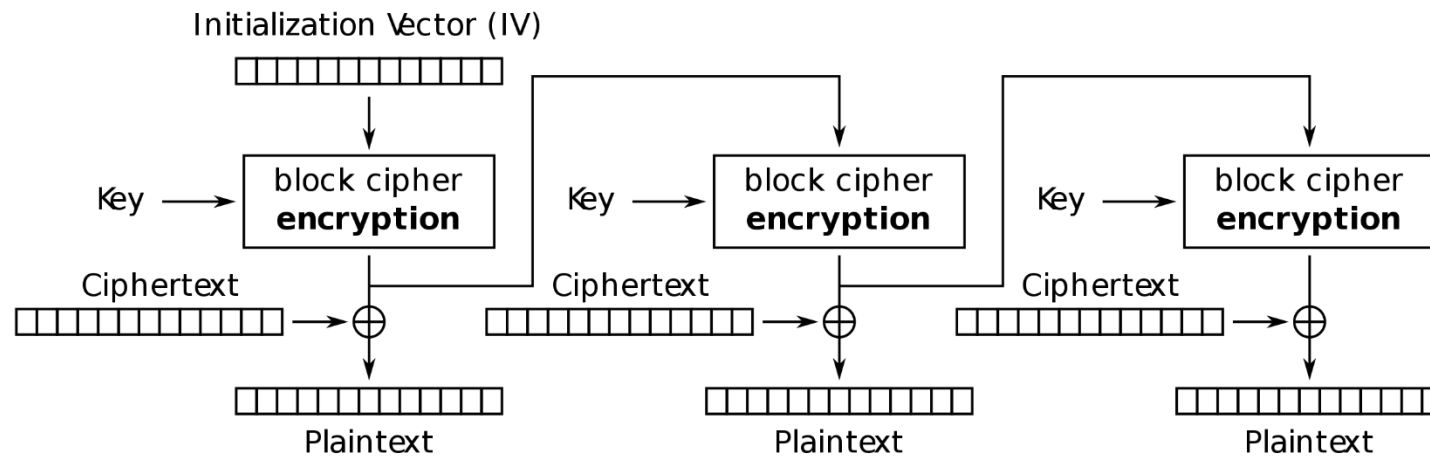


- Weaknesses:
 - CBC requires the reliable transmission of all the blocks sequentially
 - CBC is not suitable for applications that allow packet losses (e.g., music and video streaming)

Output Feedback (OFB)



Output Feedback (OFB) mode encryption



Output Feedback (OFB) mode decryption

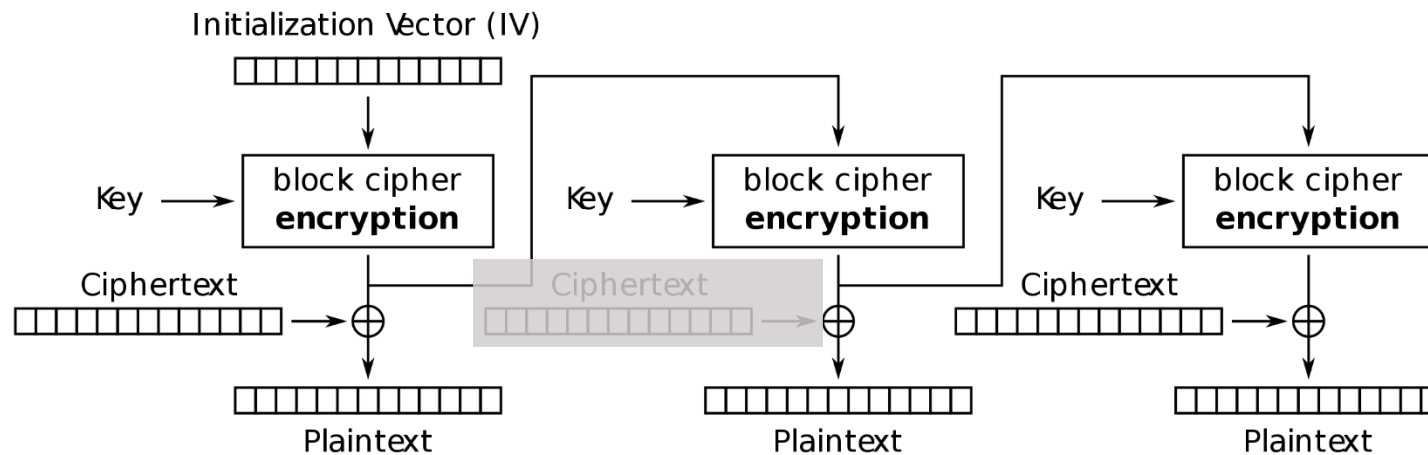
Output Feedback (OFB) modes

- Strengths:

- Can it tolerate packet loss (e.g. video streaming)?

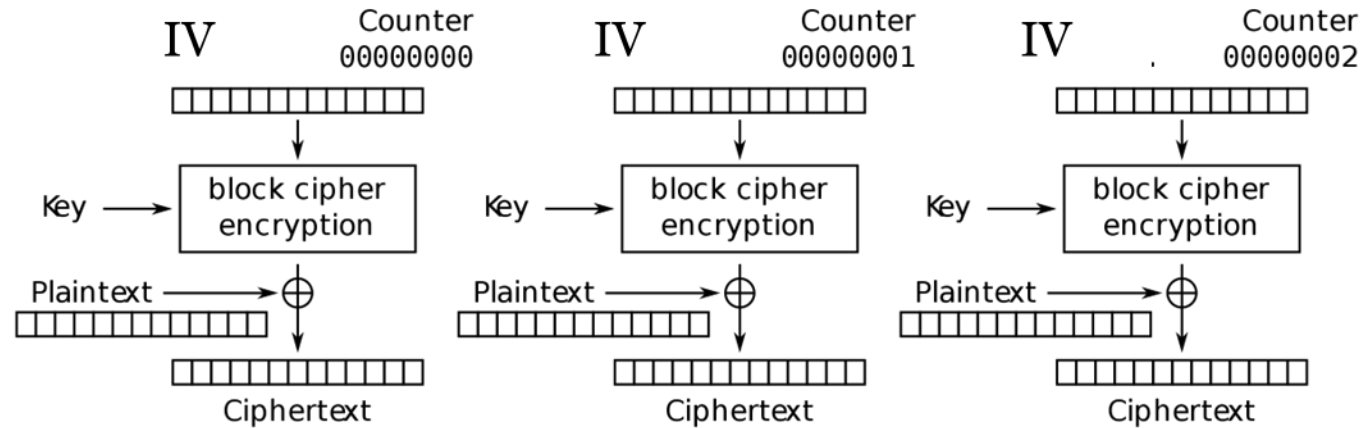
- Weaknesses:

- Is it parallelizable?

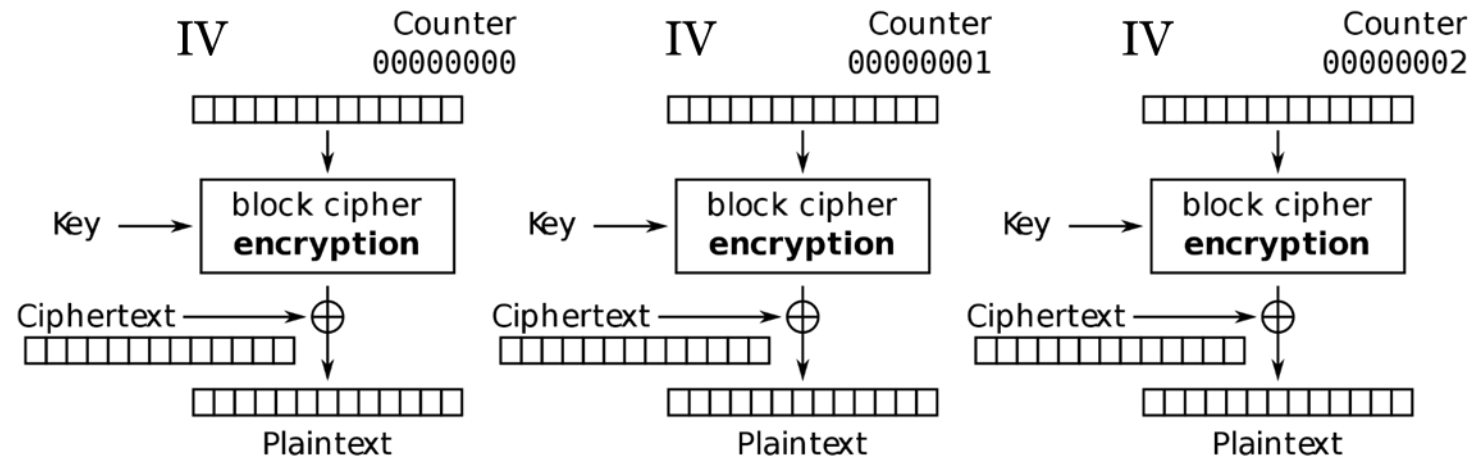


Output Feedback (OFB) mode decryption

Counter Mode (CTR)



Counter (CTR) mode encryption



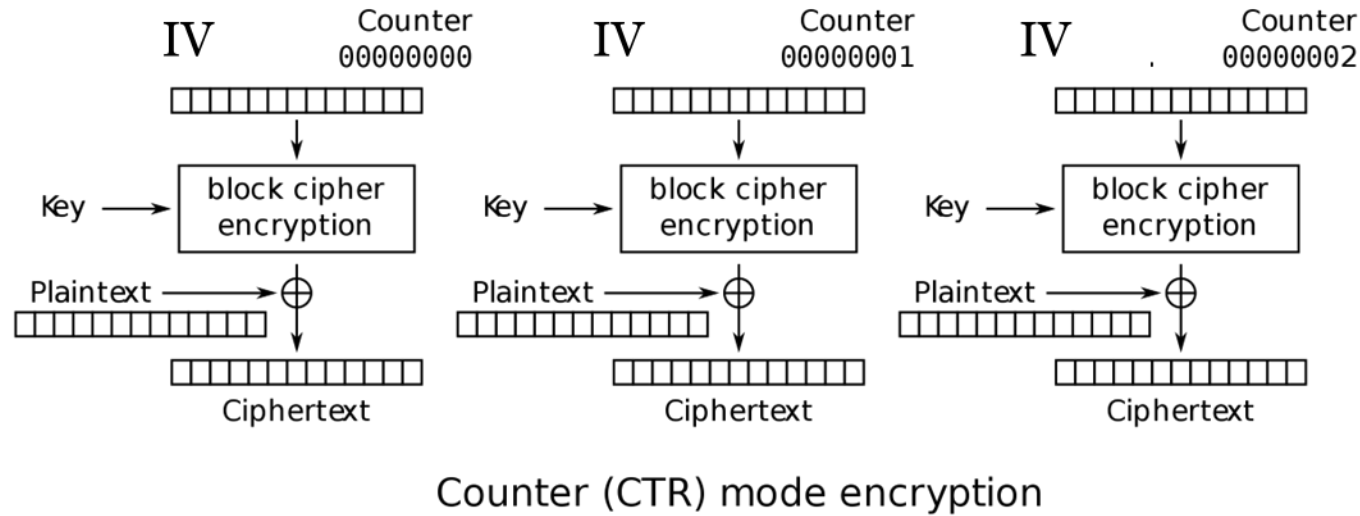
Counter (CTR) mode decryption

Counter (CTR) modes

- Strengths:
 - Can be parallelized
 - Can tolerate packet loss (e.g. streaming)
- Weaknesses:
 - be super careful about reusing the counter or IV!



Counter Mode (CTR)



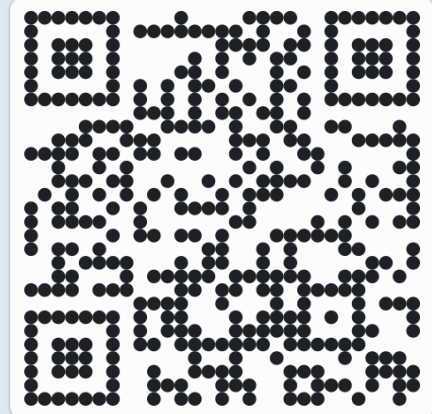
- Don't reuse IV in different encryptions
- Don't overwrap counter in one encryption
 - If counter length $n/4$, what's the maximize number of message blocks?
 - message length $\leq 2^{\frac{n}{4}}$ blocks
 - Instead, we use $IV \oplus \text{Counter}$

Theorem

If F is a PRF, then CTR mode is CPA-secure (THEOREM 3.32)

Proof:

1. Replace F with a truly random function f : difference $\leq \text{negl}(n)$
2. If $IV \oplus \text{Counter}$ s are different: $\Pr[D^{f(\cdot)}(n) = 1] = \frac{1}{2}$ one time pad and random function
3. If $IV \oplus \text{Counter}$ repeats, $\Pr[D^{f(\cdot)}(n) = 1] = 1$
4. $\Pr[IV \oplus \text{Counter repeats}] \leq \frac{2q(n)^2}{2^n}$, assuming message length $\leq 2^n$ blocks
5.
$$\Pr[D^{f(\cdot)}(n) = 1] = \Pr[D^{f(\cdot)}(n) = 1 \wedge \overline{\text{repeat}}] + \Pr[D^{f(\cdot)}(n) = 1 \wedge \text{repeat}]$$
$$\leq \Pr[D^{f(\cdot)}(n) = 1 \mid \overline{\text{repeat}}] + \Pr[\text{repeat}] = \frac{1}{2} + \frac{2q(n)^2}{2^n}$$



Chosen-plaintext attack (CPA) security

Experiment $\text{PrivK}_{A,\Pi}^{\text{CPA}}(n)$:

1. A key k is generated using $\text{Gen}(1^n)$
2. Adversary A is given input 1^n , and **oracle access to $\text{Enc}_k(\cdot)$** . A outputs two messages $m_0, m_1 \in \mathcal{M}$ s.t. $|m_0| = |m_1|$
3. A random bit $b \in \{0,1\}$ is chosen. $c = \text{Enc}_k(m_b)$ is given to A .
4. A continues to have **oracle access to $\text{Enc}_k(\cdot)$** and outputs b'
5. Experiment outputs 1 if $b=b'$, and 0 otherwise.

Definition: a private-key encryption $\Pi = (\text{Gen}, \text{Enc}, \text{Dec})$ is CPA-secure, if for all probabilistic polynomial time adversaries A , there is a negligible function negl such that for all n ,

$$\Pr[\text{PrivK}_{A,\Pi}^{\text{CPA}}(n) = 1] \leq \frac{1}{2} + \text{negl}(n)$$

Stream Cipher

- Pseudo-Random Generator PRG $G(IV, k)$
 - Pseudo-random sequence of bits $S = S[0], S[1], S[2], \dots$
- Stream cipher
 - XOR the plaintext with the key stream $C[i] = S[i] \oplus M[i]$
 - Suitable for plaintext of arbitrary length generated on the fly, e.g., media stream

Limitations of Stream Cipher

- Sequential
- Synchronization

What are the encryption modes?

× ECB mode: Electronic Codebook

- CBC mode: most commonly used
- OFB mode: packet loss
- CTR mode: parallelizable

Security- $\rightarrow$ CPA-secure



Simple Illustration of Zoom Encryption Failure



by [Davi Ottenheimer](#) on April 10, 2020

The Citizen Lab [April 3rd, 2020 report](#) broke the news on Zoom using weak encryption and gave this top-level finding:



Zoom [documentation](#) claims that the app uses “AES-256” encryption for meetings where possible. However, we find that in each Zoom meeting, a single AES-128 key is used in ECB mode by all participants to encrypt and decrypt audio and video. The use of ECB mode is not recommended because patterns present in the plaintext are preserved during encryption.

Review

Definition	Protocol	Proof	Assumption
Perfect secrecy Perfect indistinguishability	One-time pad	OTP is perfectly secure	None
Computational EAV security	PRG	proof by reduction	RC4/Chacha20 is a PRG
EAV security for multiple encryptions			
CPA security	PRF	proof by reduction	AES/DES is a PRP/PRF
	Modes of encryptions		